Designing Intelligent Agents
COMP3004
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How does different types of traders perform against each other on different numbers of traders?

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Introduction

The Bristol Stock Exchange is a simple simulation environment for a continuous double auction exchange. A continuous double auction exchange, also known as CDA, is a style of auction whereby both buyers and sellers can announce their bid and sale prices at the same time. The buyer or seller can also accept the offer available from any seller or buyer at any time. For this to be possible, a limit order book is designed to record both bids by the buyer and offers by the seller. Using this simple simulated environment, we were able to test several automated trading strategies and compare each of the effectiveness. [1]

On the Bristol Stock Exchange, several automated trading strategies are already provided for use. The strategies are Giveaway (GVWY), Insider (ISDR), Parameterized-Response Zero-Intelligence Stochastic Hill-Climber (PRSH), Parameterized-Response Zero-Intelligence (PRZI), Shaver (SHVR), Sniper (SNPR), Zero Intelligence Constrained (ZIC) and Zero Intelligence Plus (ZIP).

The first strategy, Giveaway (GVWY) is a simple algorithm which executes the price given by the customer with no intention of making any profit. The Insider (ISDR) strategy is similar to the giveaway strategy, but it does come with insider knowledge of the equilibrium price that the market will reach. The Parameterized-Response Zero-Intelligence Stochastic Hill-Climber (PRSH) strategy is a zero-intelligence strategy which is built upon the Parameterized-Response Zero-Intelligence (PRZI) which generates quote prices using a random distribution. [2] However, PRSH differentiate itself with the use of a stochastic hill climber, making it an adaptive strategy. The Shaver (SHVR) strategy uses the information on the limit order book and always cuts a penny off the best bid or offer price. Sniper (SNPR) unlike Shaver, reacts to the limit order book but with a sense of urgency. When the market is about to close, it will rapidly shave off its price with the priority to make a deal. Zero Intelligence Constrained (ZIC) strategy designed by Gode, D. K., & Sunder, S. sets a budget constraint to the automated traders as they try to prove that efficient trader requires a good structure instead of good motivation, intelligence, or learning.[4] The Zero Intelligence Plus (ZIP) strategy which was introduced afterwards by Dave Cliff, J. B. is an attempt to improve the base strategy of ZIC through the introduction of a fundamental form of machine learning.

In this project, we aim to compare and identify few of the high performing automated trading strategies where they will subsequently be chosen to be further evaluated. The evaluation will compare the automated strategies against each other using average profit per trader as the metric. The automated strategies will be put through individual environments where there will be different numbers of buyers and sellers to observe if the number of buyers and sellers affect the performance of each type of trading strategies.

Literature Review

Eric Mc Laren. worked on trading algorithms comparison on a website called Quantopian. The website provides long periods of data for trading algorithm implementation and result analysing. However, this platform trades US financial assets and it uses actual past data which is the main difference when compared to our environment of the Bristol Stock Exchange. They did advise to test the factors of the algorithms in the environment to ensure that there's a better result. [5]

Zhao, C. proposed the idea of using machine learning method to predict the stock market. However, the method in which they used to test their algorithm only chooses several stocks from the market instead of having a full simulation environment. Only four stocks are chosen, and they are Apple, Ford, Federal National Mortgage Association and Exxon Mobil Corporation. There was no simulated trading in the proposed method as well because the method uses several prediction models on the past price data. Only the final profit is compared to identify which model is more effective. [6]

Biondo, A. E., Pluchino, A., Rapisarda, A., and Helbing, D. studied whether is random trading strategies are better than technical strategies in an attempt to understand the role of randomness in financial markets. They compared 5 strategies, the Momentum, the RSI, the UPD, the MACD, and a random strategy. However, similar to the previous proposal we studied, this comparison does not involve trading simulation. It only uses the strategies to predict whether the next day will be bullish or bearish. If the prediction matches the actual data, the strategy will then get a score for that time frame. [7]

Design and Implementation

This experiment uses the Bristol Stock Exchange simulation environment to run all the traders and their experiments. [1] All the trading strategies are provided by default as part of the Bristol Stock Exchange simulation environment. However, in order for the experiments to run in sequence on a python notebook. The original BSE.py file which allows me to run the Bristol Stock Exchange environment and multiple trials of it is modified into a function which can be imported to other files. The function is also designed to be able to run with customizable parameters so that various kinds of experiments are possible through lines of codes.

The project is run on python 3.9.10 and for the two python notebook which we used, namely candidateSelection.ipynb and processResults.ipynb uses the library of pandas, string, matpotlib.pyplot, random and numpy. The BSE.py file is also imported to allow the use of Bristol Stock Exchange environment trials as part of this experiment. Both files are run using the seed of 616 to ensure that the data is replicable.

The candidateSelection.ipynb file is designed for the selection and identification of 5 highly performing automated trading strategies out of the original 8. This is mainly so that the final experiment is done on algorithms which have significant performance and to decrease the runtime of every iteration of the experiments. The candidate selection stage begins with the default setup of the Bristol Stock Exchange whereby every trading strategy is taken in with 10 buyers and 10 sellers. The experiment is run on 100 trials with every transaction recorded on the file avg_balance.csv. After the 100 trials are done, the file is read and summarised to show the average profit per trader for each trading strategy over the period of 100 trials. The data extracted is then outputted and plotted on a bar chart. The top 5 trading strategies are then chosen for the next step of evaluation.

The processResults.ipynb file is designed to run the 5 selected highly performing trading strategies on multiple experiments where there are different numbers of buyers and sellers to observe their difference in performance. The result processing stage runs a total of 14 experiments where each experiments have its own combinations of buyers and sellers. The experiments can be split into 3 categories where the first has the same number of buyer and sellers, the second has different numbers of buyers and sellers and the third and final uses a random number generator that generates a number between 1 and 20 for the number of buyers and sellers for each individual trading strategy. The first category consists of 1 buyer 1 seller, 2 buyers 2 sellers, 5 buyers 5 sellers, 10 buyers 10 sellers and 20 buyers 20 sellers. The second category has 1 buyer 5 sellers, 1 buyer 20 sellers, 2 buyers 10 sellers, 5 buyers 1 seller, 5 buyers 20 sellers, 10 buyers 2 sellers, 20 buyers 1 seller and 20 buyers 5 sellers. Each experiment is first run with the specified numbers of buyers and sellers before outputting the average profit per trader over the 100 trials in each experiment and plotting it on a graph.

Results

During the candidate selection stage, the running of the experiment of 100 trials using the default 10 buyers and 10 sellers have shown that the ISDR, SHVR and ZIP strategy has performed the best with them have an average profit per trader of 255.6, 253.4 and 248.2 respectively. This is followed by GVWY where it gets an average profit per trader of 2515.4 per trader and for the fifth place, ZIC stood out against PRSH and PRZI with an average profit per trader of 186 while the other 2 gets 172.9 and 158.6. The SNPR performed the worst with an average profit per trader of 57.6. The figure below shows the average profit per trader from each trading strategy and a bar chart containing the data. Table shows all the average profit per trader from each trading strategy.

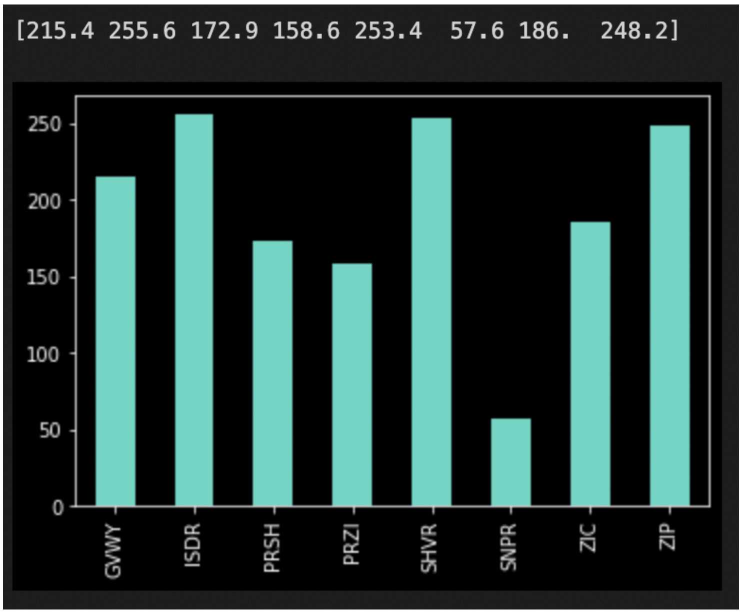


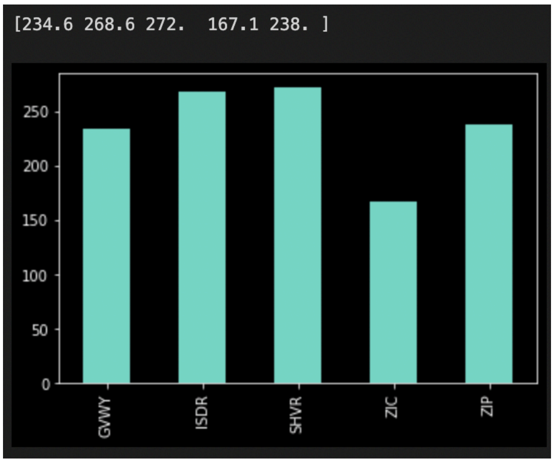
Figure 1: The average profit per trader for all 8 trading algorithm over 100 trials on the Bristol Stock Exchange with 10 buyers and sellers each. The above list shows the average profit per trader rounded to 1 decimal place and below shows the bar chart.

Numbers of Buyer and Seller	GVWY	ISDR	PRSH	PRZI	SHVR	SNPR	ZIC	ZIP
10 Buyer, 10 Seller	**215.4**	**255.6**	172.9	158.6	**253.4**	57.6	**186**	**248.2**

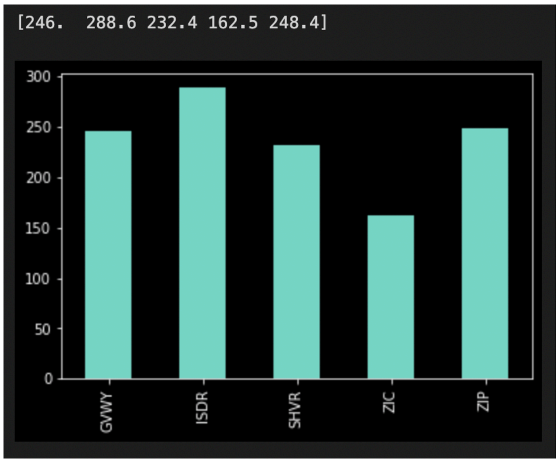
Table 1: The average profit per trader for all 8 trading strategy over 100 trials on the Bristol Stock Exchange with 10 buyers and sellers each. Highlighted in black are those that got chosen for final evaluation

The candidate selection stage has shown that GVWY, ISDR, SHVR, ZIC and ZIP strategy are the highest performing strategy when using average profit per trader as the comparison metric. Therefore, these 5 strategies are chosen for the next stage for the comparison of performance when there are different numbers of traders.

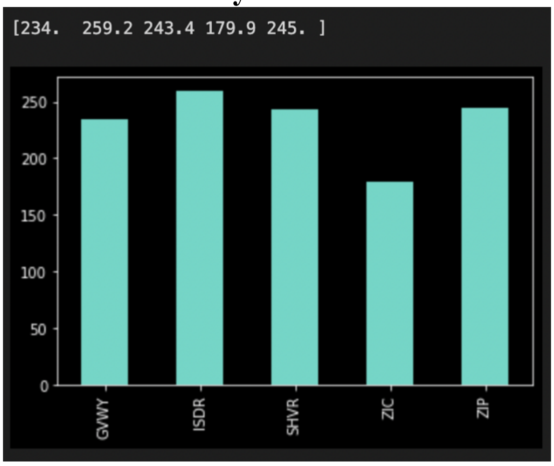
In the experiments for the final evaluations, there are 3 categories as stated in the previous section. In the first category where both buyer and seller have the same numbers, there are a total of 5 experiments. Throughout these 5 experiments, ISDR and SHVR has shown promising result with both getting the highest average profit per trader in 2 experiments. ISDR got 288.6 in 2 buyers 2 sellers and 259.2 in 5 buyers 5 sellers while SHVR got 272 in 1 buyer 1 seller and 255.9 in 10 buyers 10 sellers. The other strategy that is left with a highest average profit per trader is ZIP where it achieves 257.4 when there is 20 buyers and 20 sellers. GVWY and ZIC did not get any achievements in terms of highest average profit per trader in all 5 experiments. However, if we look at the strategy that got the highest average profit per trader in all five experiments, ISDR gets the crown with the average profit per trader of 288.6 in the experiment with 2 buyers 2 sellers. All the average profit per trader produced by all trading strategies in these 5 experiments is between 162.5 and 288.6.



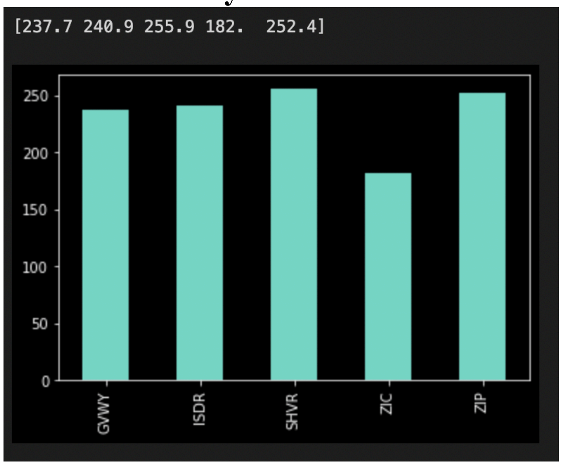
1 buyer 1 seller



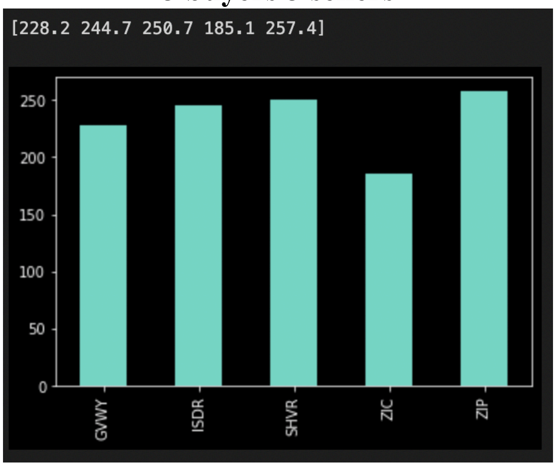
2 buyers 2 sellers



5 buyers 5 sellers



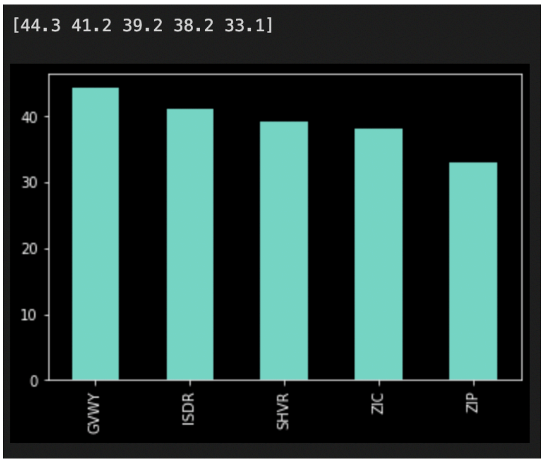
10 buyers 10 sellers



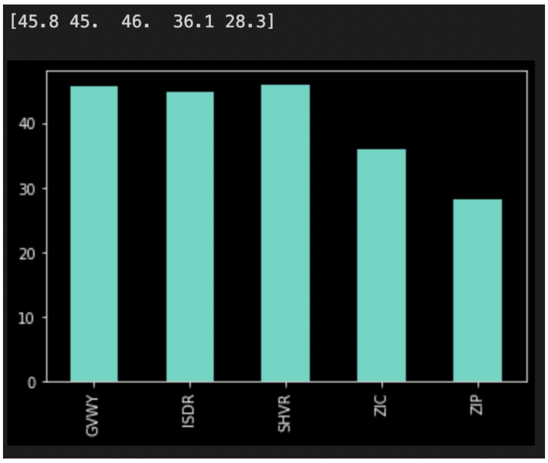
20 buyers 20 sellers

Figure 2 to 6 (From left to right, top to bottom): The average profit per trader for all 5 trading strategies over 100 trials on the Bristol Stock Exchange with different numbers of traders. The above list shows the average profit per trader rounded to 1 decimal place and below shows the bar chart.

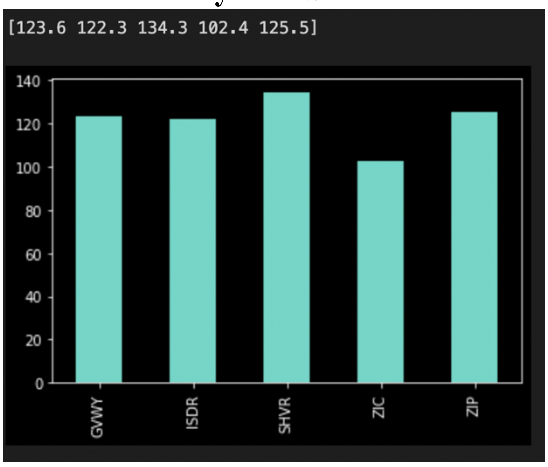
In the second category where both buyer and seller are in different numbers, there are a total of 8 experiments. Within these 8 experiments, SHVR performs the best with it getting the highest average profit per trader in 3 experiments. The experiments are when there's 20 buyers 1 seller with average profit per trader of 46, 1 buyer 10 sellers with average profit per trader of 134.3 and 5 buyers 20 sellers with average profit per trader of 162.7. The second place is tied between GVWY and ZIP strategies where both get highest average profit per trader in 2 experiments. GVWY gets an average profit per trader of 44.3 in 1 buyer 20 sellers and 136 in 10 buyers 2 sellers while ZIP achieved 135.6 and 159.5 of average profit per trader for 1 buyer 5 sellers and 20 buyers 5 sellers respectively. ISDR did get highest average profit per trader but only in one experiment when there are 5 buyers and 1 seller. The average profit per trader achieved in that experiment is 132.5. ZIC trading strategy is the only strategy that did not achieve highest average profit per trader in all 8 experiments. 162.7 is the highest average profit per trader result from all 8 experiments and it is produced by the SHVR strategy in 5 buyers 20 sellers. The range of average profit per trader by all trading strategies in these 8 experiments is between 28.3 and 162.7.



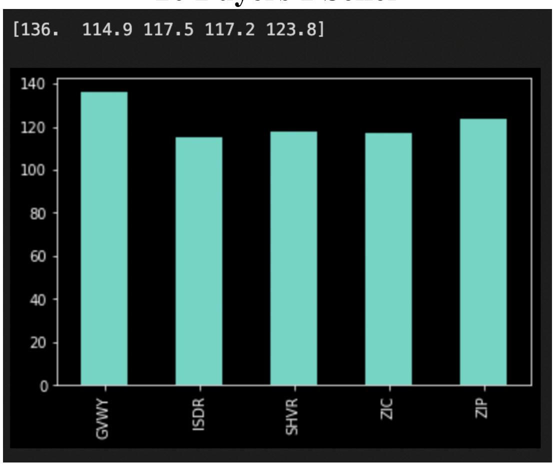
1 Buyer 20 Sellers



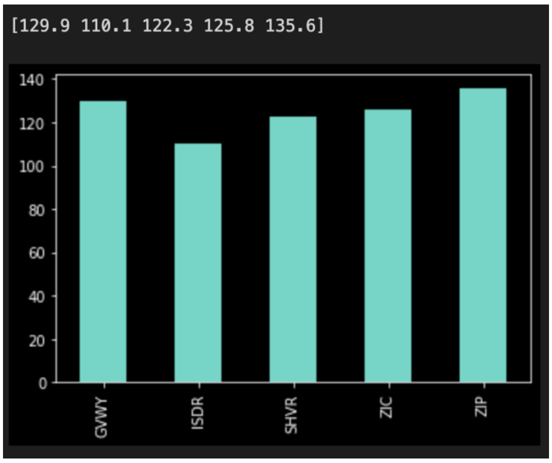
20 Buyers 1 Seller



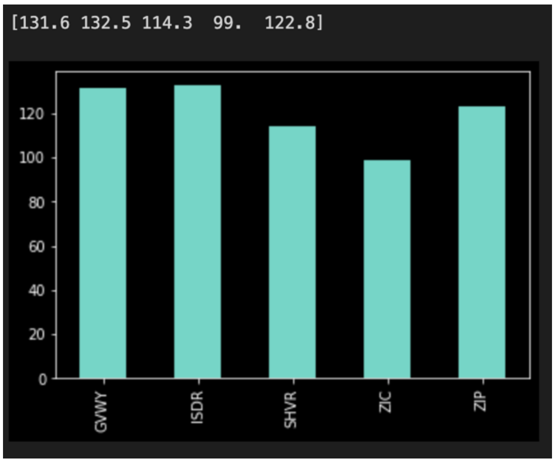
2 Buyers 10 Sellers



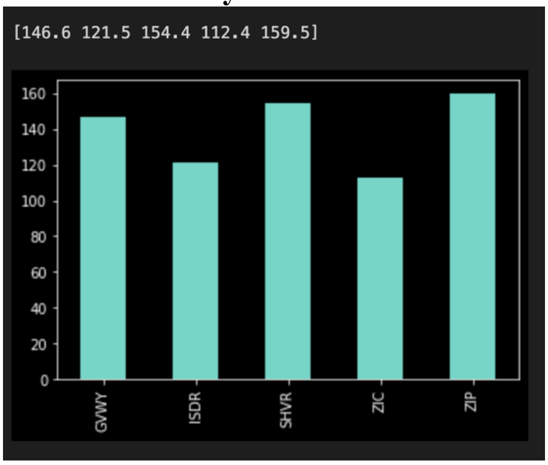
10 Buyers 2 Sellers



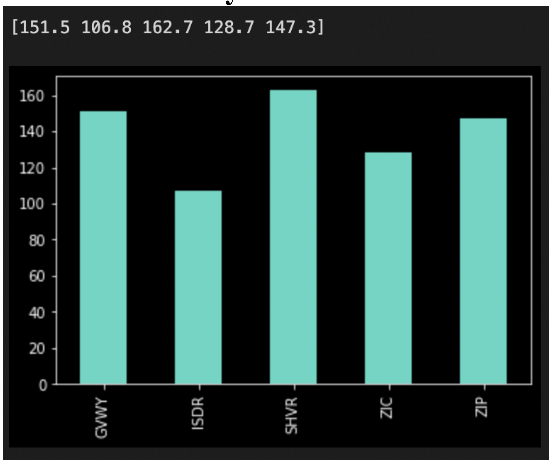
1 Buyer 5 Sellers



5 Buyers 1 Seller



20 Buyers 5 Sellers



5 Buyers 20 Sellers

Figure 7 to 14 (From left to right, top to bottom): The average profit per trader for all 5 trading strategies over 100 trials on the Bristol Stock Exchange with different numbers of traders. The above list shows the average profit per trader rounded to 1 decimal place and below shows the bar chart.

For the final category, there are only one experiment which uses a random number generator for the numbers of buyers and sellers for all 5 trading strategies. The numbers generated are 1 buyer 9 sellers for GVWY, 15 buyers 13 sellers for ISDR, 17 buyers 4 sellers for SHVR, 6 buyers 12 sellers for ZIC and 16 buyers 5 sellers for ZIP. The resulting average profit per trader for all 5 trading strategies are 313.4 for GVWY, 75.3 for ISDR, 192.6 for SHVR, 211.3 for ZIC and 198.9 for ZIP. The highest average profit per trader is achieved by GVWY and the lowest is gotten by ISDR. The range of average profit per trader in this experiment is between 75.3 and 313.4.

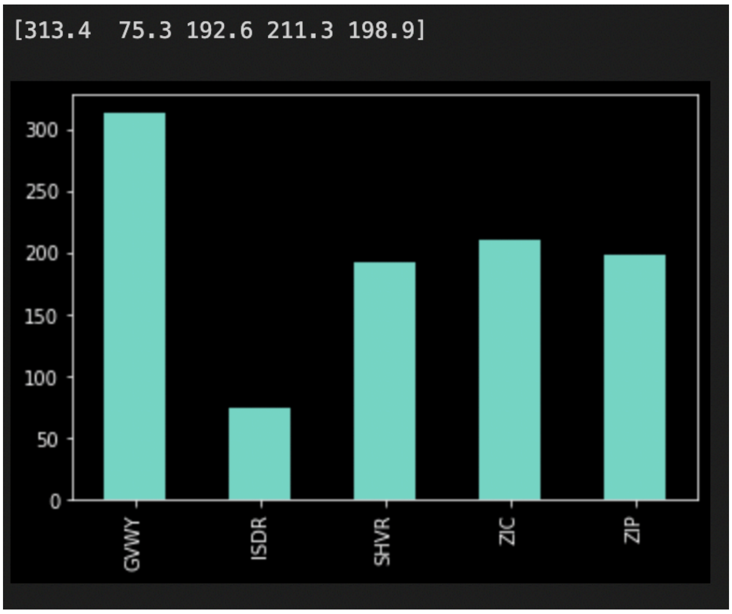


Figure 15: The average profit per trader for all 5 trading strategies over 100 trials on the Bristol Stock Exchange with randomly generated numbers of traders. The above list shows the average profit per trader rounded to 1 decimal place and below shows the bar chart.

Overall, throughout 14 experiments, SHVR strategy achieve the highest average profit per trader the most times by getting it in 4 experiments. GVWY, ISDR and ZIP shares the second place by getting such achievements in 3. ZIC strategy is the only one which did not get the highest average profit per trader in any experiments. The highest average profit per trader in all 14 experiments is achieved by GVWY with the profit of 313.4 in the final experiment where it has 1 buyer and 9 sellers. The lowest is achieved by ZIP with the profit of 28.3 in the experiment with 20 buyers and 1 seller. The table below summarises all the resulting average profit per trader for all 5 trading strategies over the 14 experiments which uses different numbers of buyers and sellers.

Numbers of Buyer and Seller	GVWY	ISDR	SHVR	ZIC	ZIP
Category 1: Same Buyer and Seller Number					
1 Buyer, 1 Seller	234.6	268.6	**272**	167.1	238
2 Buyers, 2 Sellers	246	**288.6**	232.4	162.5	248.4
5 Buyers, 5 Sellers	234	**259.2**	243.4	179.9	245
10 Buyers, 10 Sellers	237.7	240.9	**255.9**	182	252.4
20 Buyers, 20 Sellers	228.2	244.7	250.7	185.1	**257.4**
Category 2: Different Buyer and Seller Number					
1 Buyer, 20 Sellers	**44.3**	41.2	39.2	38.2	33.1
20 Buyers, 1 Seller	45.8	45	**46**	36.1	28.3
2 Buyers, 10 Sellers	123.6	122.3	**134.3**	102.4	125.5
10 Buyers, 2 Sellers	**136**	114.9	117.5	117.2	123.8
1 Buyer, 5 Sellers	129.9	110.1	122.3	125.8	**135.6**
5 Buyers, 1 Seller	131.6	**132.5**	114.3	99	122.8
20 Buyers, 5 Sellers	146.6	121.5	154.4	112.4	**159.5**
5 Buyers, 20 Sellers	151.5	106.8	**162.7**	128.7	147.3
Category 3: Random Buyer and Seller Number					
Random No. Buyer, Random No. Seller	**313.4**	75.3	192.6	211.3	198.9

Table 2: The average profit per trader for the 5 selected trading strategy over 100 trials on the Bristol Stock Exchange in different combinations of numbers of buyers and sellers. Highlighted are those which achieved the highest number of average profits per trader in that experiment.

Discussion

Through observations on the results above, we have made several interesting discoveries. Starting with experiments in category 1, it shows that while the number of traders increases exponentially from 1 buyer and 1 seller to 20 buyer and 20 sellers in the fifth experiment, there is no significant changes to the average profit per trader for all 5 trading strategies. There are also no significant trendline which indicates either an increase or decrease of the average profit per trader as the number of both buyer and seller increases at the same time.

Experiments in category 2 does show interesting results especially when we look at the experiments of 1 buyer 20 seller and 20 buyers and 1 seller. This significant difference between number of buyers and number of sellers has kept the average profit per trader between the range of 28.3 to 46 which is a huge decrease from the range of 162.5 and 288.6 throughout all experiments in category 1 when there are no differences between the numbers of buyer and seller in each experiment. However, when we lower the difference between number of buyer and seller from 19 to 15, the average profit per trader increases exponentially to the range between 106.8 and 151.5. The average profit per trader range then remains in a similar range even when the difference between buyer and seller drop to 8 and 4. Experiments from the category 2 has shown that minor differences in the number of buyers and sellers such as 4 would decrease the average profit per trader around half and major differences beginning at 19 would further cut the number by a third. This also indicates that the wider the differences between the number of buyers and sellers, the lower the average profit per trader across all 5 different trading strategies. The results from category 2 also does not show any significant trendline that indicates that the number difference between buyer and seller would affect the average profit per trader of any of the 5 trading strategies individually.

Category 3 while only has 1 experiment, is the only experiment which has different numbers of buyers and sellers for every single trading strategy. This provides the only result of different numbers of buyers and sellers of different trading strategies interacting with each other which has produces interesting results which includes highest average profit per trader of 313.4 from GVWY even though it has only a difference between buyer and seller of 8 and a low average profit per trader of 75.3 even though the difference between buyer and seller is 2. This suggests that while a lower difference between buyer and seller should increase the average profit per trader, the same may not apply when all trading strategies have different numbers of buyer and seller. This subject can be further studied in a future work where a proper method of experiment is needed to explore to identify the pattern and trends in the experiment.

Conclusion

This project which uses a minimal trading environment simulation has enabled the exploration the differences in performance of several high performing trading strategies when they have different numbers of buyer and seller. The project has shown that low differences between number of buyer and seller would produce high average profit per trader and as the differences increases, the average profit per trader will also decrease. However, this only works when all the trading strategies have the same buyer and seller number. If all trading strategies have different number of buyer and seller, the interaction between the traders would be different thus the resulting average profit per trader will be different. This subject can be further explored in future works where a framework of experiments is designed to produce data compatible for pattern and trends identification.

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